BiasAperture — Demographic Fairness Audit Dossier

Diagnostic Demographic Bias Evaluation & Regulatory Traceability Report

AUDITED MODEL
MobileNetV2 0.35 Rep Gender Classifier (Seeed SSCMA)

EVALUATION DATASET
FairFace Benchmark Dataset (Validation Split) (N=10954)

PROTECTED AXIS
Race

AUDIT DATE
2026-10-16 04:59:28 UTC

Diagnostic Scope & Non-Intervention Guarantee:

BiasAperture is engineered exclusively as an external diagnostic auditing, statistical significance testing, and regulatory traceability platform. It does **not** perform model retraining, in-processing debiasing, or synthetic image generation. In adherence to statistical integrity safeguards (NFR-003), demographic subgroups with fewer than 30 samples ($n < 30$) have point estimates suppressed to prevent unreliable compliance claims.

3 / 7	Black	EOP = 0.080	0
SUBGROUPS WITH SIGNIFICANT DISPARITY (P < 0.05)	LARGEST DEVIATION SUBGROUP	WORST HEADLINE METRIC	SUBGROUPS FLAGGED N < 30 (NFR-003)

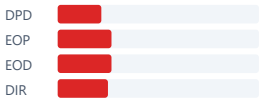
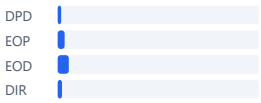
1. Headline Fairness Metrics (Core Four — Cross-Group Overview)

Summary metrics evaluated across all eligible demographic strata (n ≥ 30 per NFR-003). Accompanied by 95% stratified BCa bootstrap confidence intervals and Pearson's χ^2 test of independence. *Note: each CI band is scaled to its own row — dot colour (red = CI excludes null), not bar width, indicates significance across rows.*

Metric Name	Point Estimate	95% Bootstrap CI	CI Band vs. Null	Significance (p-value)	Regulatory Clause
Demographic Parity Difference	0.125	[0.099, 0.155]		p=0.0000 p < 0.05	EU AI Act Art. 10(2)(f) / NIST Measure 2.11
Equalized Odds Difference	0.137	[0.102, 0.160]		p=0.0000 p < 0.05	EU AI Act Art. 10(2)(g) / NIST Measure 2.11
Equal Opportunity Difference	0.137	[0.100, 0.161]		p=0.0000 p < 0.05	EU AI Act Art. 10(3) / NIST Measure 2.11
Disparate Impact Ratio	0.855	[0.824, 0.885]		p=0.0000 p < 0.05	EEOC 4/5ths Rule / EU AI Act Annex IV §2(g)

2. Per-Subgroup Disparity Matrix

Detailed stratification across individual demographic groups, ordered by largest deviation from the null (no-disparity) point, most significant first. Subgroups below the NFR-003 sample threshold (n < 30) are strictly flagged, uncomputed to prevent small-sample estimation artifacts, and summarised separately below.

Subgroup	Sample Size (n)	DPD	EOD (Max-Gap)	EOP	DIR (Symmetric)	Deviation from Null (red = p < 0.05)	Data Integrity Status
Black	1556	0.065	0.080	0.080	0.925		Audited (Valid)
Middle Eastern	1209	0.060	0.059	0.051	0.925		Audited (Valid)
White	2085	0.042	0.057	0.057	0.948		Audited (Valid)
Southeast Asian	1415	0.018	0.034	0.034	0.978		Audited (Valid)
Indian	1516	0.021	0.025	0.025	0.974		Audited (Valid)
East Asian	1550	0.005	0.020	0.007	0.994		Audited (Valid)
Latino_Hispanic	1623	0.005	0.017	0.010	0.994		Audited (Valid)

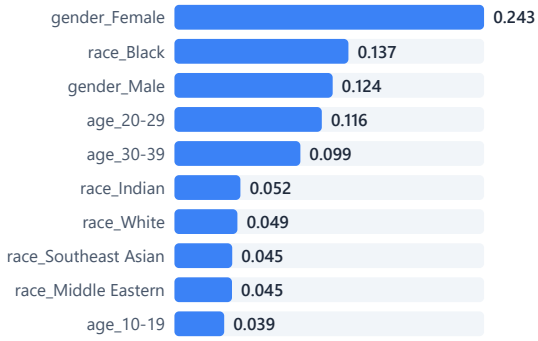
3. Targeted Disparity Explainability & Proxy Attributions

Surrogate additive Shapley attributions computed for statistically flagged disparities ($p < 0.05$, $n \geq 30$). Bar lengths represent mean absolute Shapley attribution magnitude across the evaluation cohort. Only the top proxy features are shown.

> **Non-Causal Disclaimer (Bilodeau et al., 2022):** Attributions identify empirical associations between demographic proxy features and model predictions. They do *not* establish causal relationships or prescribe remediation. This is a diagnostic tool only — no model retraining or weight modification is performed.

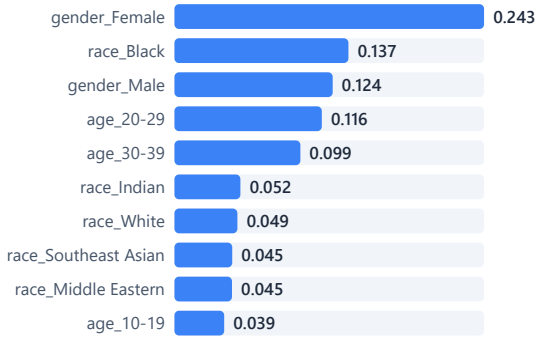
Flagged Disparity: `demographic_parity_difference` for subgroup ALL

Surrogate Shapley attribution computed across 10954 subjects for ALL. These demographic surrogate associations summarize prediction log-odds across the full audit; they do not explain images, causation, or the disparity itself.



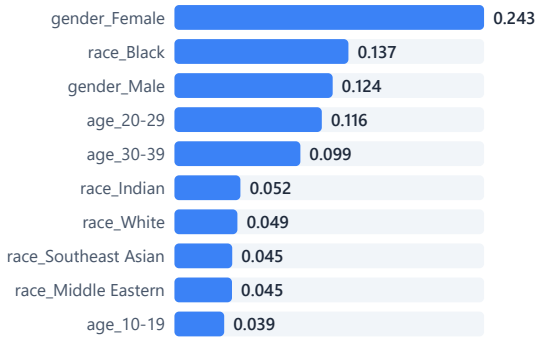
Flagged Disparity: `equalized_odds_difference` for subgroup ALL

Surrogate Shapley attribution computed across 10954 subjects for ALL. These demographic surrogate associations summarize prediction log-odds across the full audit; they do not explain images, causation, or the disparity itself.



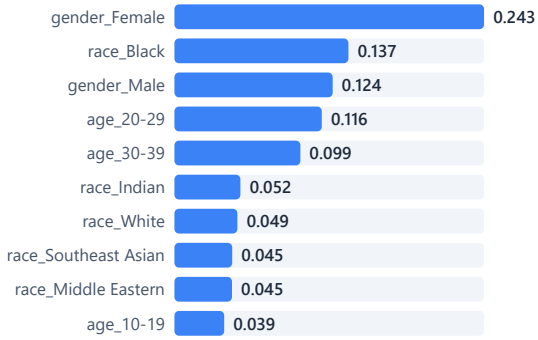
Flagged Disparity: `equal_opportunity_difference` for subgroup ALL

Surrogate Shapley attribution computed across 10954 subjects for ALL. These demographic surrogate associations summarize prediction log-odds across the full audit; they do not explain images, causation, or the disparity itself.



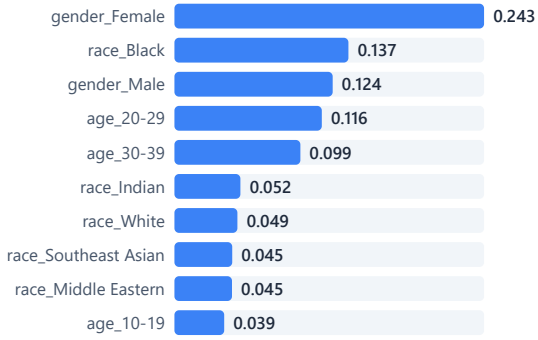
Flagged Disparity: `disparate_impact_ratio` for subgroup ALL

Surrogate Shapley attribution computed across 10954 subjects for ALL. These demographic surrogate associations summarize prediction log-odds across the full audit; they do not explain images, causation, or the disparity itself.



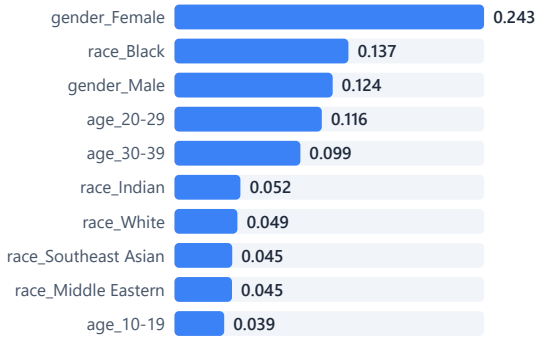
Flagged Disparity: `demographic_parity_difference` for subgroup Black

Surrogate Shapley attribution computed across 10954 subjects for Black. These demographic surrogate associations summarize prediction log-odds across the full audit; they do not explain images, causation, or the disparity itself.



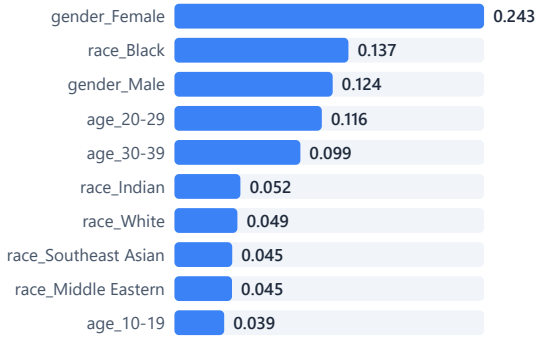
Flagged Disparity: `equal_opportunity_difference` for subgroup Black

Surrogate Shapley attribution computed across 10954 subjects for Black. These demographic surrogate associations summarize prediction log-odds across the full audit; they do not explain images, causation, or the disparity itself.



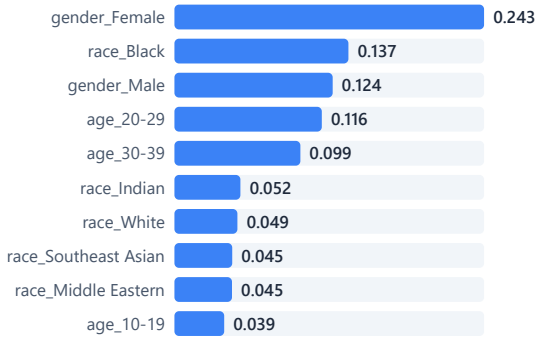
Flagged Disparity: `equalized_odds_difference` for subgroup Black

Surrogate Shapley attribution computed across 10954 subjects for Black. These demographic surrogate associations summarize prediction log-odds across the full audit; they do not explain images, causation, or the disparity itself.



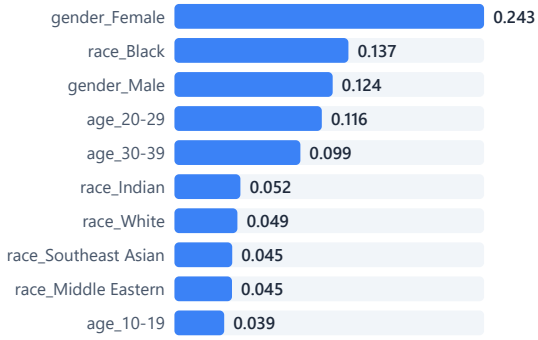
Flagged Disparity: `disparate_impact_ratio` for subgroup Black

Surrogate Shapley attribution computed across 10954 subjects for Black. These demographic surrogate associations summarize prediction log-odds across the full audit; they do not explain images, causation, or the disparity itself.



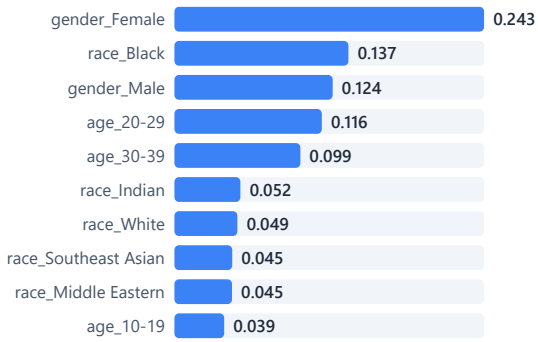
Flagged Disparity: `demographic_parity_difference` for subgroup Middle Eastern

Surrogate Shapley attribution computed across 10954 subjects for Middle Eastern. These demographic surrogate associations summarize prediction log-odds across the full audit; they do not explain images, causation, or the disparity itself.



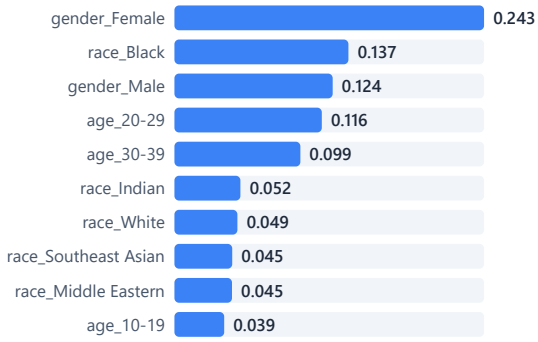
Flagged Disparity: `equal_opportunity_difference` for subgroup Middle Eastern

Surrogate Shapley attribution computed across 10954 subjects for Middle Eastern. These demographic surrogate associations summarize prediction log-odds across the full audit; they do not explain images, causation, or the disparity itself.



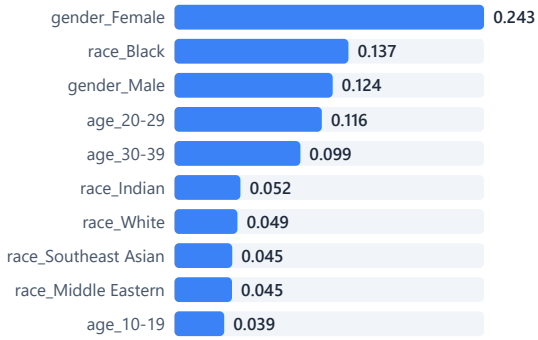
Flagged Disparity: `equalized_odds_difference` for subgroup Middle Eastern

Surrogate Shapley attribution computed across 10954 subjects for Middle Eastern. These demographic surrogate associations summarize prediction log-odds across the full audit; they do not explain images, causation, or the disparity itself.



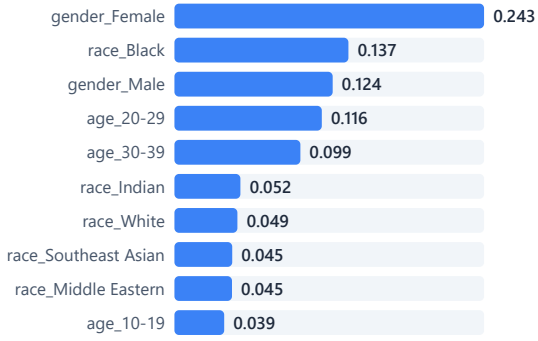
Flagged Disparity: `disparate_impact_ratio` for subgroup Middle Eastern

Surrogate Shapley attribution computed across 10954 subjects for Middle Eastern. These demographic surrogate associations summarize prediction log-odds across the full audit; they do not explain images, causation, or the disparity itself.



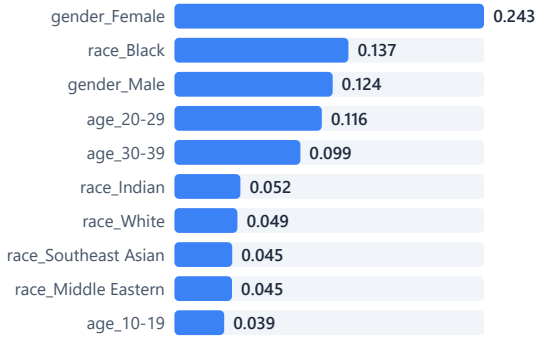
Flagged Disparity: `demographic_parity_difference` for subgroup White

Surrogate Shapley attribution computed across 10954 subjects for White. These demographic surrogate associations summarize prediction log-odds across the full audit; they do not explain images, causation, or the disparity itself.



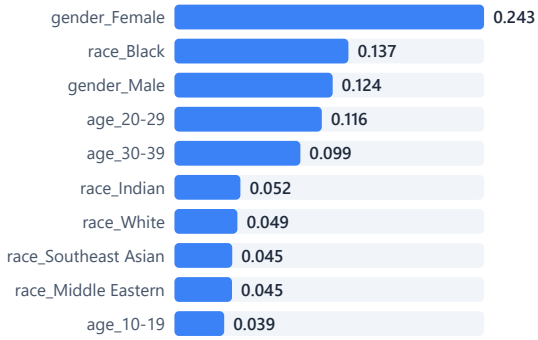
Flagged Disparity: `equal_opportunity_difference` for subgroup White

Surrogate Shapley attribution computed across 10954 subjects for White. These demographic surrogate associations summarize prediction log-odds across the full audit; they do not explain images, causation, or the disparity itself.



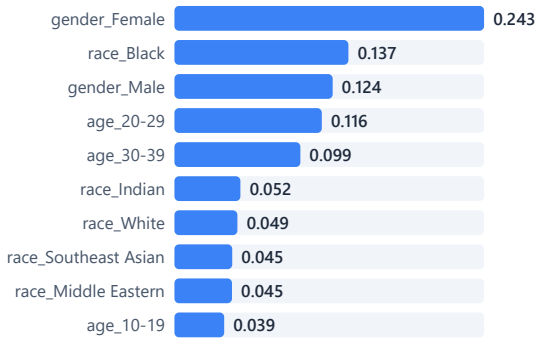
Flagged Disparity: `equalized_odds_difference` for subgroup White

Surrogate Shapley attribution computed across 10954 subjects for White. These demographic surrogate associations summarize prediction log-odds across the full audit; they do not explain images, causation, or the disparity itself.



Flagged Disparity: `disparate_impact_ratio` for subgroup White

Surrogate Shapley attribution computed across 10954 subjects for White. These demographic surrogate associations summarize prediction log-odds across the full audit; they do not explain images, causation, or the disparity itself.



Demographic Surrogate Associations

These associations describe a surrogate fitted to predictions across the full audit. They do not establish causation or explain individual images or disparity metrics. Image-native SHAP is deferred.

ALL — demographic_parity_difference

Surrogate Shapley attribution computed across 10954 subjects for ALL. These demographic surrogate associations summarize prediction log-odds across the full audit; they do not explain images, causation, or the disparity itself.

Demographic feature	Mean absolute contribution to surrogate log-odds
gender_Female	0.2433
race_Black	0.1367
gender_Male	0.1243
age_20-29	0.1158
age_30-39	0.0989
race_Indian	0.0518
race_White	0.0495
race_Southeast Asian	0.0454
race_Middle Eastern	0.0452
age_10-19	0.0391

ALL — equalized_odds_difference

Surrogate Shapley attribution computed across 10954 subjects for ALL. These demographic surrogate associations summarize prediction log-odds across the full audit; they do not explain images, causation, or the disparity itself.

Demographic feature	Mean absolute contribution to surrogate log-odds
gender_Female	0.2433
race_Black	0.1367
gender_Male	0.1243
age_20-29	0.1158
age_30-39	0.0989
race_Indian	0.0518
race_White	0.0495

Demographic feature	Mean absolute contribution to surrogate log-odds
race_Southeast Asian	0.0454
race_Middle Eastern	0.0452
age_10-19	0.0391

ALL — equal_opportunity_difference

Surrogate Shapley attribution computed across 10954 subjects for ALL. These demographic surrogate associations summarize prediction log-odds across the full audit; they do not explain images, causation, or the disparity itself.

Demographic feature	Mean absolute contribution to surrogate log-odds
gender_Female	0.2433
race_Black	0.1367
gender_Male	0.1243
age_20-29	0.1158
age_30-39	0.0989
race_Indian	0.0518
race_White	0.0495
race_Southeast Asian	0.0454
race_Middle Eastern	0.0452
age_10-19	0.0391

ALL — disparate_impact_ratio

Surrogate Shapley attribution computed across 10954 subjects for ALL. These demographic surrogate associations summarize prediction log-odds across the full audit; they do not explain images, causation, or the disparity itself.

Demographic feature	Mean absolute contribution to surrogate log-odds
gender_Female	0.2433
race_Black	0.1367
gender_Male	0.1243
age_20-29	0.1158
age_30-39	0.0989
race_Indian	0.0518

Demographic feature	Mean absolute contribution to surrogate log-odds
race_White	0.0495
race_Southeast Asian	0.0454
race_Middle Eastern	0.0452
age_10-19	0.0391

Black — demographic_parity_difference

Surrogate Shapley attribution computed across 10954 subjects for Black. These demographic surrogate associations summarize prediction log-odds across the full audit; they do not explain images, causation, or the disparity itself.

Demographic feature	Mean absolute contribution to surrogate log-odds
gender_Female	0.2433
race_Black	0.1367
gender_Male	0.1243
age_20-29	0.1158
age_30-39	0.0989
race_Indian	0.0518
race_White	0.0495
race_Southeast Asian	0.0454
race_Middle Eastern	0.0452
age_10-19	0.0391

Black — equal_opportunity_difference

Surrogate Shapley attribution computed across 10954 subjects for Black. These demographic surrogate associations summarize prediction log-odds across the full audit; they do not explain images, causation, or the disparity itself.

Demographic feature	Mean absolute contribution to surrogate log-odds
gender_Female	0.2433
race_Black	0.1367
gender_Male	0.1243
age_20-29	0.1158
age_30-39	0.0989

Demographic feature	Mean absolute contribution to surrogate log-odds
race_Indian	0.0518
race_White	0.0495
race_Southeast Asian	0.0454
race_Middle Eastern	0.0452
age_10-19	0.0391

Black — equalized_odds_difference

Surrogate Shapley attribution computed across 10954 subjects for Black. These demographic surrogate associations summarize prediction log-odds across the full audit; they do not explain images, causation, or the disparity itself.

Demographic feature	Mean absolute contribution to surrogate log-odds
gender_Female	0.2433
race_Black	0.1367
gender_Male	0.1243
age_20-29	0.1158
age_30-39	0.0989
race_Indian	0.0518
race_White	0.0495
race_Southeast Asian	0.0454
race_Middle Eastern	0.0452
age_10-19	0.0391

Black — disparate_impact_ratio

Surrogate Shapley attribution computed across 10954 subjects for Black. These demographic surrogate associations summarize prediction log-odds across the full audit; they do not explain images, causation, or the disparity itself.

Demographic feature	Mean absolute contribution to surrogate log-odds
gender_Female	0.2433
race_Black	0.1367
gender_Male	0.1243
age_20-29	0.1158

Demographic feature	Mean absolute contribution to surrogate log-odds
age_30-39	0.0989
race_Indian	0.0518
race_White	0.0495
race_Southeast Asian	0.0454
race_Middle Eastern	0.0452
age_10-19	0.0391

Middle Eastern — demographic_parity_difference

Surrogate Shapley attribution computed across 10954 subjects for Middle Eastern. These demographic surrogate associations summarize prediction log-odds across the full audit; they do not explain images, causation, or the disparity itself.

Demographic feature	Mean absolute contribution to surrogate log-odds
gender_Female	0.2433
race_Black	0.1367
gender_Male	0.1243
age_20-29	0.1158
age_30-39	0.0989
race_Indian	0.0518
race_White	0.0495
race_Southeast Asian	0.0454
race_Middle Eastern	0.0452
age_10-19	0.0391

Middle Eastern — equal_opportunity_difference

Surrogate Shapley attribution computed across 10954 subjects for Middle Eastern. These demographic surrogate associations summarize prediction log-odds across the full audit; they do not explain images, causation, or the disparity itself.

Demographic feature	Mean absolute contribution to surrogate log-odds
gender_Female	0.2433
race_Black	0.1367

Demographic feature	Mean absolute contribution to surrogate log-odds
gender_Male	0.1243
age_20-29	0.1158
age_30-39	0.0989
race_Indian	0.0518
race_White	0.0495
race_Southeast Asian	0.0454
race_Middle Eastern	0.0452
age_10-19	0.0391

Middle Eastern — equalized_odds_difference

Surrogate Shapley attribution computed across 10954 subjects for Middle Eastern. These demographic surrogate associations summarize prediction log-odds across the full audit; they do not explain images, causation, or the disparity itself.

Demographic feature	Mean absolute contribution to surrogate log-odds
gender_Female	0.2433
race_Black	0.1367
gender_Male	0.1243
age_20-29	0.1158
age_30-39	0.0989
race_Indian	0.0518
race_White	0.0495
race_Southeast Asian	0.0454
race_Middle Eastern	0.0452
age_10-19	0.0391

Middle Eastern — disparate_impact_ratio

Surrogate Shapley attribution computed across 10954 subjects for Middle Eastern. These demographic surrogate associations summarize prediction log-odds across the full audit; they do not explain images, causation, or the disparity itself.

Demographic feature	Mean absolute contribution to surrogate log-odds
gender_Female	0.2433
race_Black	0.1367
gender_Male	0.1243
age_20-29	0.1158
age_30-39	0.0989
race_Indian	0.0518
race_White	0.0495
race_Southeast Asian	0.0454
race_Middle Eastern	0.0452
age_10-19	0.0391

White — demographic_parity_difference

Surrogate Shapley attribution computed across 10954 subjects for White. These demographic surrogate associations summarize prediction log-odds across the full audit; they do not explain images, causation, or the disparity itself.

Demographic feature	Mean absolute contribution to surrogate log-odds
gender_Female	0.2433
race_Black	0.1367
gender_Male	0.1243
age_20-29	0.1158
age_30-39	0.0989
race_Indian	0.0518
race_White	0.0495
race_Southeast Asian	0.0454
race_Middle Eastern	0.0452
age_10-19	0.0391

White — equal_opportunity_difference

Surrogate Shapley attribution computed across 10954 subjects for White. These demographic surrogate associations summarize prediction log-odds across the full audit; they do not explain images, causation, or the disparity itself.

Demographic feature	Mean absolute contribution to surrogate log-odds
gender_Female	0.2433
race_Black	0.1367
gender_Male	0.1243
age_20-29	0.1158
age_30-39	0.0989
race_Indian	0.0518
race_White	0.0495
race_Southeast Asian	0.0454
race_Middle Eastern	0.0452
age_10-19	0.0391

White — equalized_odds_difference

Surrogate Shapley attribution computed across 10954 subjects for White. These demographic surrogate associations summarize prediction log-odds across the full audit; they do not explain images, causation, or the disparity itself.

Demographic feature	Mean absolute contribution to surrogate log-odds
gender_Female	0.2433
race_Black	0.1367
gender_Male	0.1243
age_20-29	0.1158
age_30-39	0.0989
race_Indian	0.0518
race_White	0.0495
race_Southeast Asian	0.0454
race_Middle Eastern	0.0452
age_10-19	0.0391

White — disparate_impact_ratio

Surrogate Shapley attribution computed across 10954 subjects for White. These demographic surrogate associations summarize prediction log-odds across the full audit; they do not explain images, causation, or the disparity itself.

Demographic feature	Mean absolute contribution to surrogate log-odds
gender_Female	0.2433
race_Black	0.1367
gender_Male	0.1243
age_20-29	0.1158
age_30-39	0.0989
race_Indian	0.0518
race_White	0.0495
race_Southeast Asian	0.0454
race_Middle Eastern	0.0452
age_10-19	0.0391

Governance Documentation & Intended Use

> **EU AI Act (Regulation EU 2024/1689) Article 10 / Annex IV Alignment:** This audit dossier provides verifiable diagnostic documentation of demographic performance disparities. It does not perform model retraining or in-situ weight modifications.

Model Description: ImageNet-pretrained ResNet-34 terminating in an 18-unit linear head sliced for race (7), gender (2), and age (9) classification.

Backend configuration: dual. Both backend executions completed; 0 point-estimate discrepancies detected.

Cross-backend agreement checks point estimates only; bootstrap, hypothesis tests, and reporting share implementation. Consult execution logs for backend failures and divergences.